Real-Time Project Research Project

# PERFORMANCE ANALYSIS OF A HEXAGONAL TWISTED CAPILLARY TUBE IN REFRIGERATORS


Presented by:

Tangirala Vamsi Krishna 22011P2103

2nd Year B.Tech IDP (MTE)

*Under the Esteemed Guidance of*

**Dr. Ravi Gugulothu**

**Assistant Professor (Contract)**

JNTUH UNIVERSITY COLLEGE OF ENGINEERING, SCIENCE AND TECHNOLOGY, HYDERABAD
Department of Mechanical Engineering

# Introduction

- Expansion Device:
  - Essential for separating high and low-pressure sides in refrigeration systems
  - Positioned between the receiver and evaporator, reducing high-pressure liquid refrigerant to low-pressure

- Capillary Tubes:
  - Most commonly used throttling devices in refrigeration systems
  - Indirectly influence the amount of refrigerant consumed
  - Straight and helical-shaped tubes are widely used designs

- Refrigerants:
  - Absorb and release heat, facilitating temperature regulation
  - Work by cycling through phases of evaporation and condensation
  - The efficiency of refrigerants in heat absorption and release determines the effectiveness of cooling systems

# Literature Review

| YEAR | AUTHOR | EXPERIMENTAL CONDITIONS/ PARAMETERS | OUTCOMES |
|---|---|---|---|
| 2023 | Ashraf Mimi Elsaid et al. | numerically investigate the performance of shell and helically coiled tube heat exchangers with circular, elliptic, and square tube cross-sections using single nanofluids by considering inclination angle, helically coiled tube (HCT) diameter ratio, flow direction, cross-section geometry, nanomaterials | observed that the circular tube cross-sections exhibit the highest Nusselt number i.e. 62.5% greater than others |
| 2022 | Ahmad Zarei et al. | experimentally explored enhancing cold thermal energy storage by using bubble injection in a helical coil heat exchanger, and its impact on Nusselt number, storage tank temperature differences, exergy degradation, and cycle coefficient of performance | achieved optimal results for a flow rate of 9 L/min. They developed empirical correlation as a function of bubble flow rate and bubble injection angle to calculate the Nusselt number in the helical coil tube. |
| 2021 | Kamel Sigar Hmood | analyzed the operational performance of small-scale refrigeration systems by examining the outlet parameters of their capillary tubes, which are crucial for accurate pressure and vapor quality determination | Results show vapor quality ranging from 0.2124 (for R600) to 0.3132 (for R1234yf), with relative errors between experimental and model results for outlet pressure and temperature ranging from 1.16% to 5.35% and 0.12% to 0.81%, respectively |
| 2019 | A. Mokhtari Ardekani et al. | investigated fluid flow and heat transfer characteristics of nanofluids (Ag and $SiO_2$/water) flowing through helically coiled tubes under uniform heat flux and Turbulent flow conditions for various dimensions of the tube | Helically coiled tubes can improve the heat transfer rate |

# Contd…

| YEAR | AUTHOR | EXPERIMENTAL CONDITIONS/ PARAMETERS | OUTCOMES |
|---|---|---|---|
| 2018 | Dubba Santhosh Kumar et al. | investigated the performance of a helically coiled capillary tube with adiabatic flow of R-600a | They found that the straight tubes outperform coiled ones by 1.5 to 16%. |
| 2017 | Pravin Jadhav et al. | investigated the flow characteristics of a helical coiled adiabatic capillary tube for transcritical $CO_2$ | developed a model for two-phase flow, analyzed it with different friction factor materials |
| 2016 | Andrew Michael Fsadni et al. | examined the pressure drop characteristics of two-phase flow in a helically coiled tube, for various scenarios like steam-water, R-134a evaporation and condensation, air-water flow, and nanofluid flow | provided researchers with a practical summary of the relevant correlations and supporting theory for calculating the two-phase pressure drop in helically coiled tubes |
| 2016 | Masoud Zareh et al. | conducted numerical simulations and experiments on R12, R22, and R134a flow through straight and helical capillary tubes under adiabatic conditions using drift flux model | Findings show that mass flux increases with higher degrees of sub-cooling, coiled diameter, and tube diameter. Additionally, void fraction and vapor quality variations exhibit similar trends in both tube configurations |
| 2014 | Hossien Shokouhmand et al. | experimentally validated choked refrigerant flow through straight and helical capillary tubes using R-134a with a 5.5% average error | Key findings enhance understanding of refrigerant flow dynamics, aiding refrigeration and air conditioning optimization |

# Contd…

| YEAR | AUTHOR | EXPERIMENTAL CONDITIONS/ PARAMETERS | OUTCOMES |
|---|---|---|---|
| 2011 | Sukkarin Chingulpitak et al. | numerically analyzed the flow behaviors of helical and straight capillary tubes | They observed that the straight tubes exhibit slightly lower pressure drop when compared to helical ones, and the helical tubes with a 40 mm coil diameter produced 9% decrease in mass flow rate when compared to straight tubes. |
| 2010 | Sukkarin Chingulpitak et al. | experimentally studied the flow characteristics of alternative refrigerants, examined them using different helical tube geometries using R-22 and other refrigerants, and compared them with the theoretical data | They found that traditional refrigerants result in shorter capillary lengths and lower pressure drops compared to alternative refrigerants. They noted that variation in coil diameter, being less than 300 mm, significantly affects tube length, whereas pitch variations, for more than 300 mm do not have any impact. |
| 2010 | Mittal M.K. et al. | investigated the coiling effect on the flow of R-407C in adiabatic helical capillary tubes | coiling reduces the mass flow rate by 5-10% compared to straight tubes |

# Motivation

- **Environmental Impact of Refrigerants**:
  - Traditional refrigerants release chlorofluorocarbons (CFCs), which damage the ozone layer.
  - Modern refrigerants, while ozone-friendly, are potent greenhouse gases contributing to global warming.

- **Energy Efficiency Concerns**:
  - Inefficient heat transfer through capillary tube leads to increased cycling of the refrigerator.
  - More cycles result in higher power consumption, impacting both the environment and energy costs.

- **Solution Through Design Optimization**:
  - By optimizing the design of capillary tubes, heat transfer efficiency can be maximized.
  - This optimization can reduce the number of cycles and, consequently, the power consumption of refrigerators

# Objectives

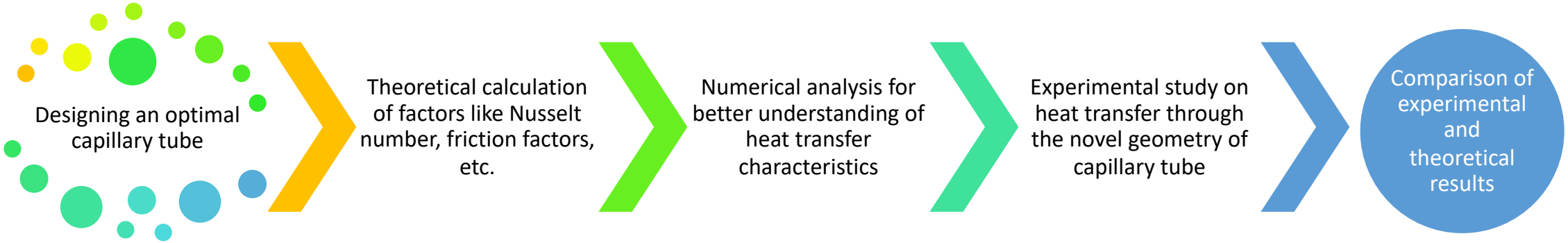
- **Maximizing heat transfer efficiency**:
  - Design geometries optimization.
  - Enhancing flow characteristics to ensure better heat exchange.

- Reduce power consumption by reducing number of compressor cycles

- Prolong system longevity by reducing strain on compressor and other components

- Augmentation of Coefficient of Performance for various capillary tubes

# Methodology



# Design Parameters

- Geometric Configuration:
  - Straight Tube: Simple cylindrical shape with uniform diameter.
  - Elliptical Tube: Cylindrical with an elliptical cross-section.
  - Plus-Shaped Twisted Tube: Cylindrical with a cross-section resembling a plus sign, twisted along its length.
  - Hexagonal Shaped Twisted Tube: Cylindrical with a hexagonal cross-section, twisted along its length.

- Dimensional Specifications:
  - Circular diameter: 100mm for all types.
  - Length: 100mm for all types.

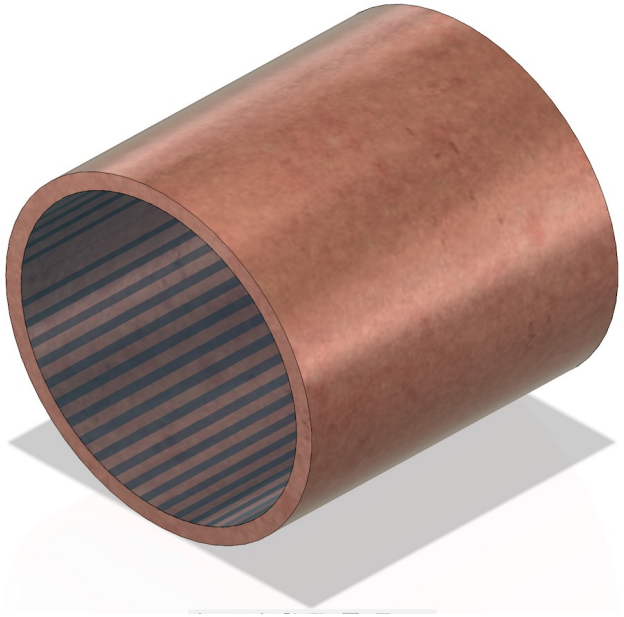


Fig: Straight Capillary Tube

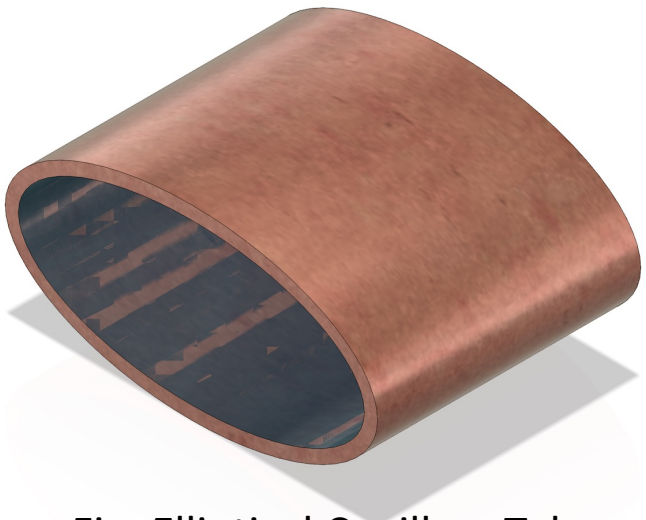


Fig: Elliptical Capillary Tube

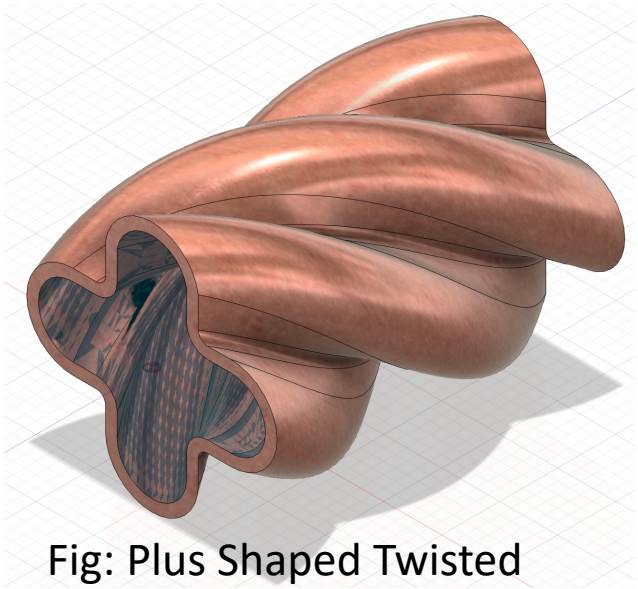


Fig: Plus Shaped Twisted Capillary Tube

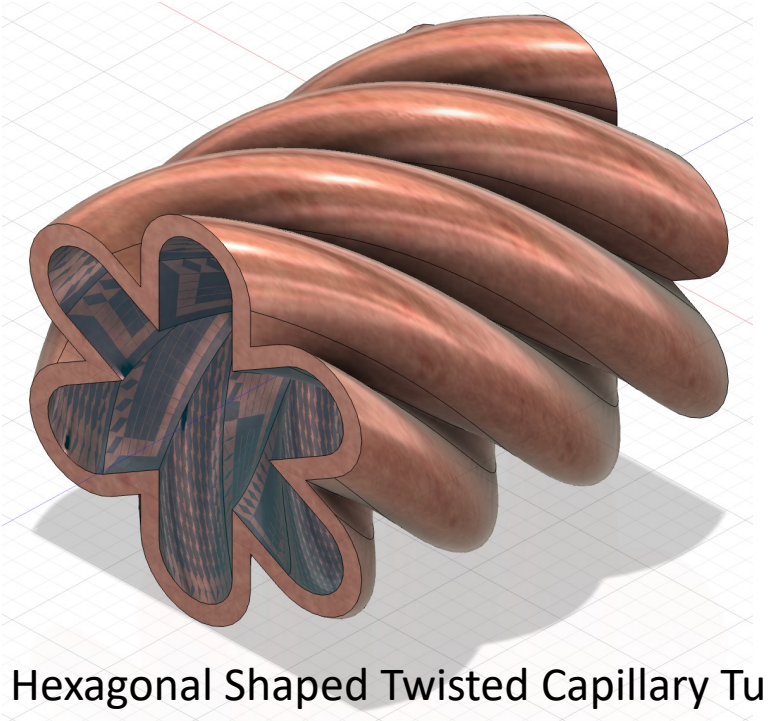


Fig: Hexagonal Shaped Twisted Capillary Tube

# Computational Simulation

- Computational analysis on all 4 designs has been performed on FUSION 360 software.

- The following are the assumed boundary conditions:
  - The ambient temperature is considered as 25ºC.
  - The static fluid temperature was assumed to be 30ºC to analyze the heat flow and heat transfer efficiently.
  - Actual flow of refrigerant in a capillary tube being turbulent, the heat transfer constant was assumed to be 5000.00 W / (m^2 K) on the inner surface of the capillary tubes
  - Assuming the external surface-air convection to be a forced convection, the heat constant is taken as 50 W / (m^2 K)
  - The radiation emissivity of copper is taken as 0.8.

# Temperature Contours

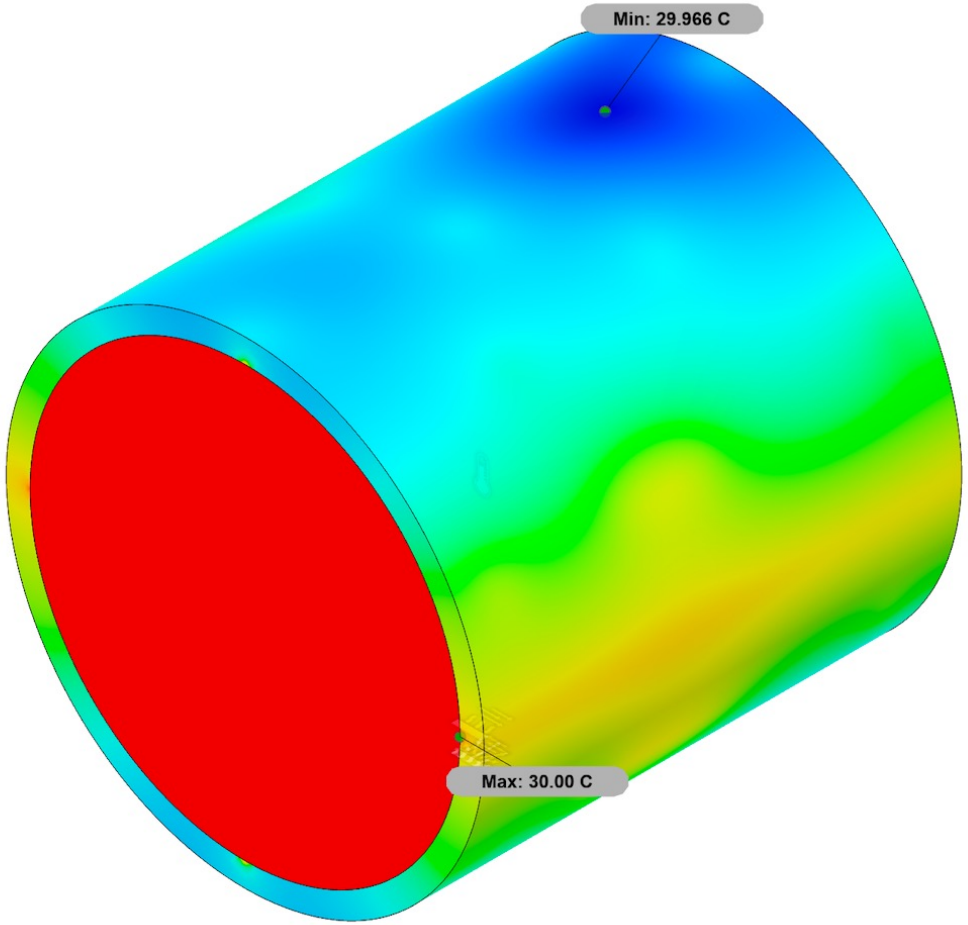



Fig: Temperature Contour of straight capillary tube
Max Temperature: 30.00ºC
Min Temperature: 29.966ºC

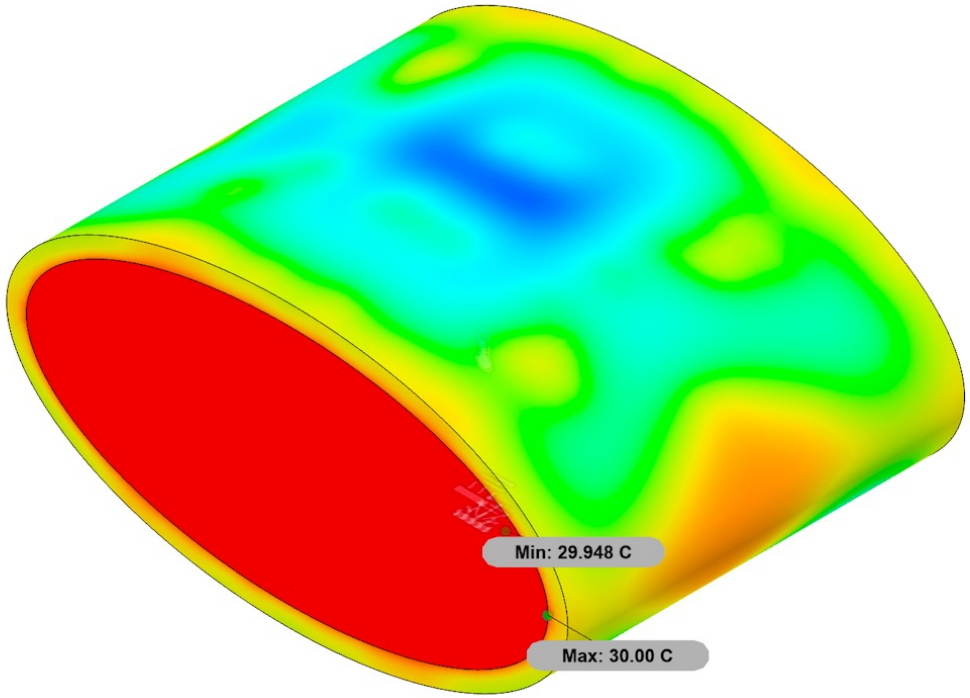



Fig: Temperature Contour of elliptical capillary tube
Max Temperature: 30.00ºC
Min Temperature: 29.948ºC

# Contd…

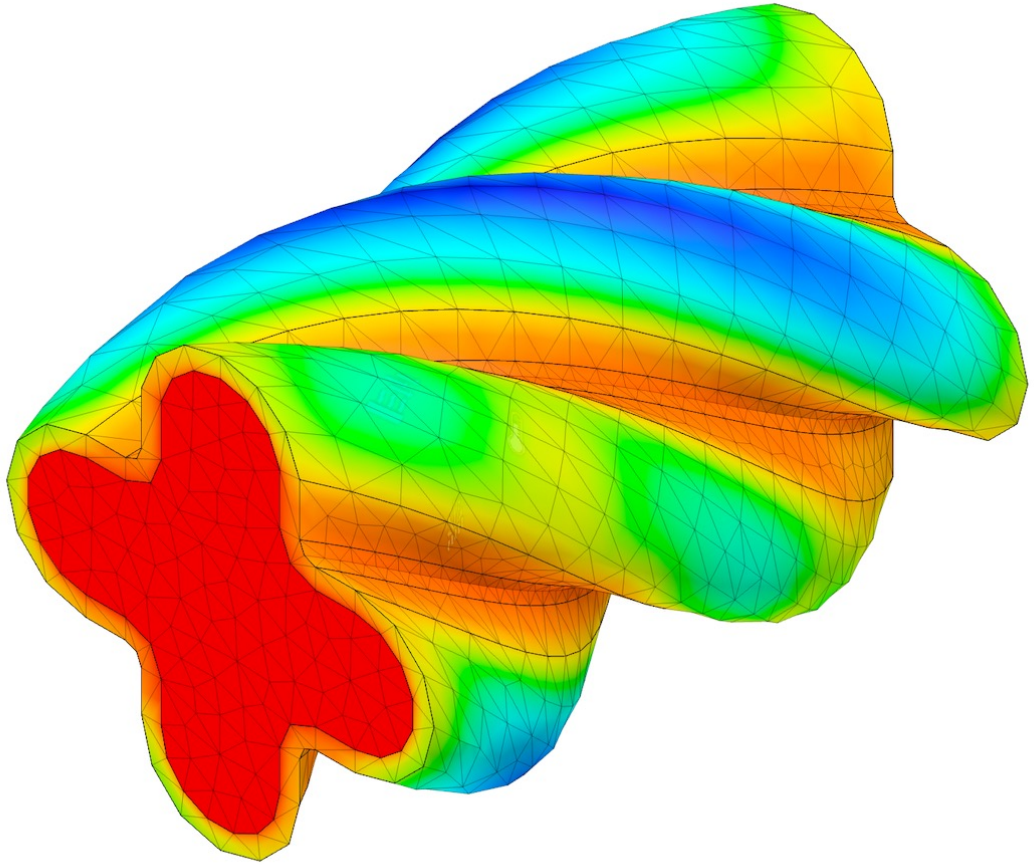


Fig: Temperature Contour of Plus shaped twisted capillary tube
Max Temperature: 30.00ºC
Min Temperature: 29.939ºC

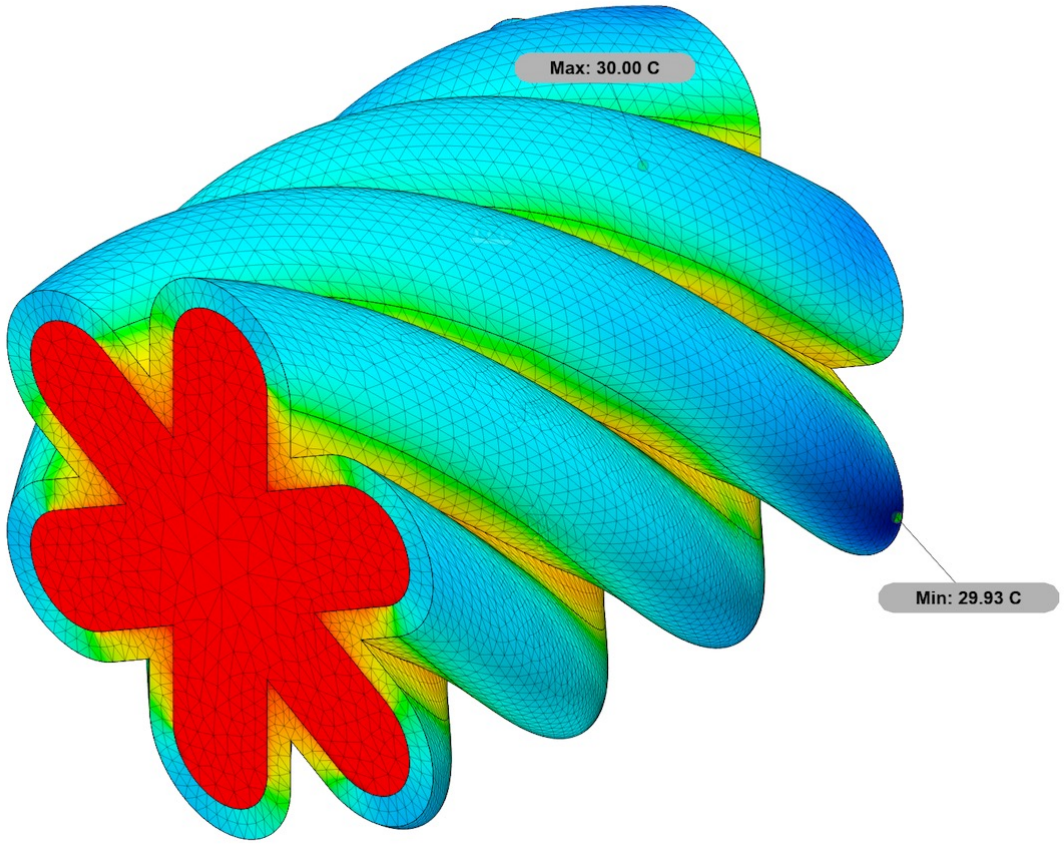
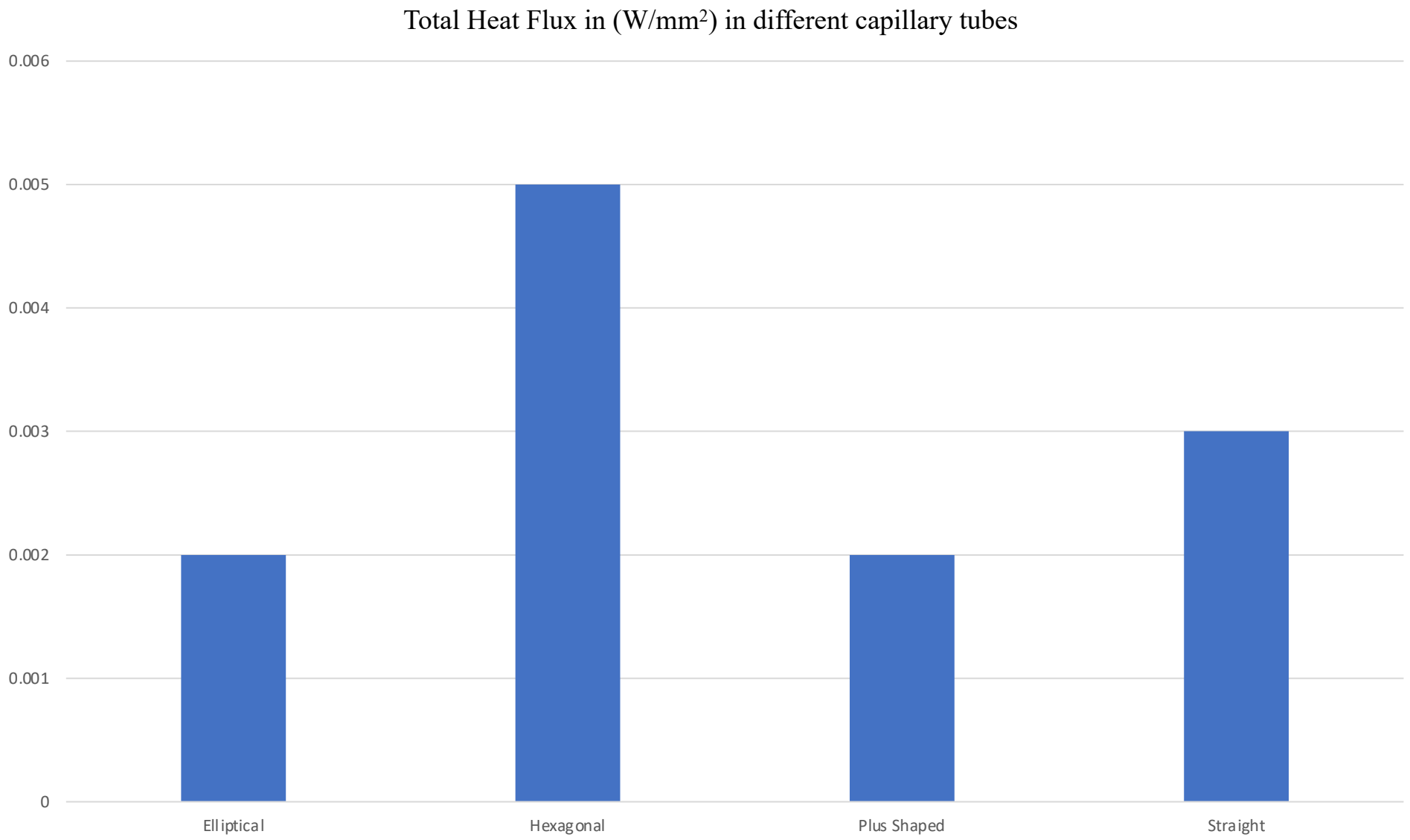
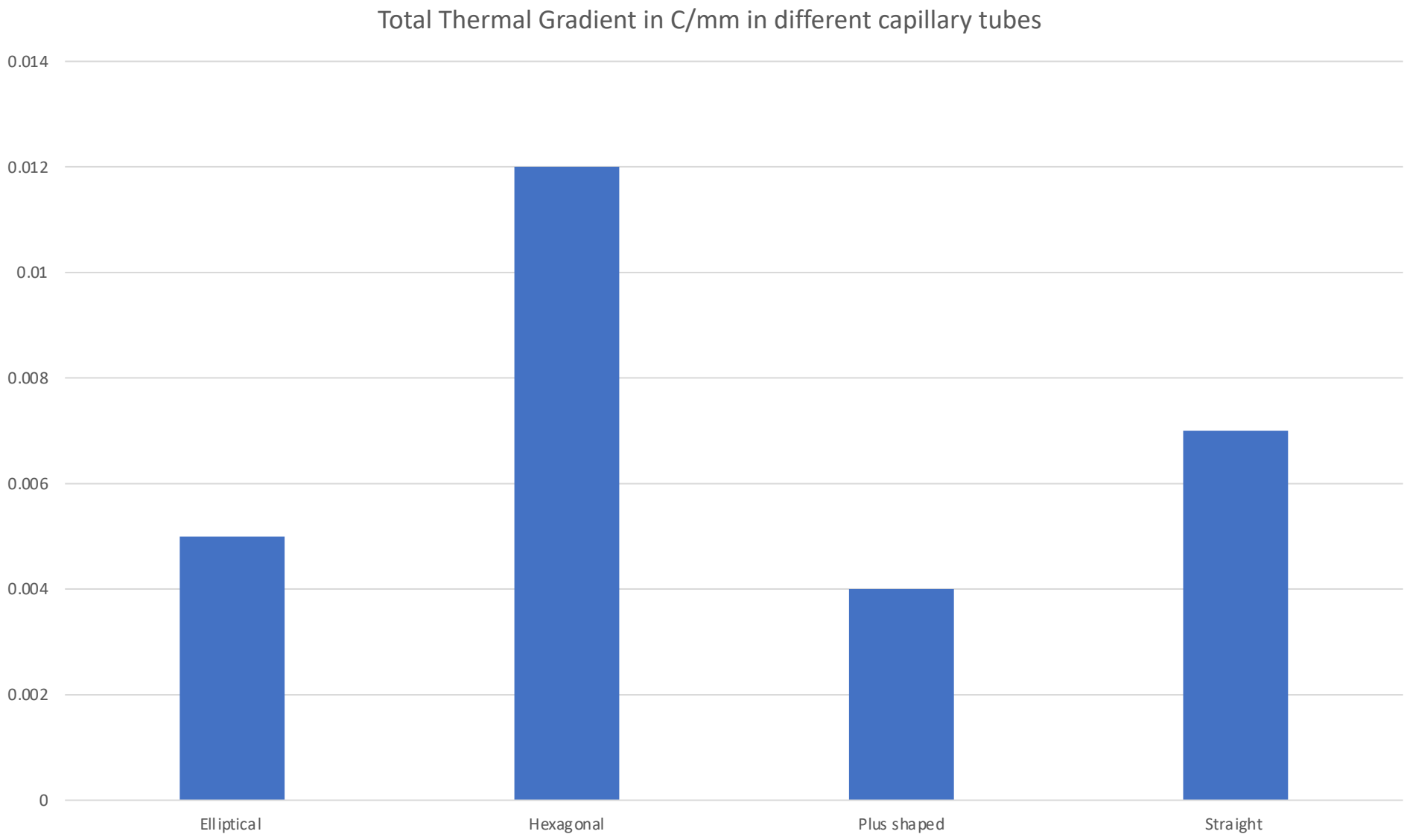



Fig: Temperature Contour of Hexagonal shaped twisted capillary tube
Max Temperature: 30.00ºC
Min Temperature: 29.93ºC

# Result


Total Heat Flux in (W/mm²) in different capillary tubes
0.006
0.005
0.004
0.003
0.002
0.001
0
Elliptical
Hexagonal
Plus Shaped
Straight

# Contd…


Total Thermal Gradient in C/mm in different capillary tubes
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
Elliptical
Hexagonal
Plus shaped
Straight

# Conclusions

- **Heat Transfer**: The hexagonal shaped twisted capillary tube demonstrates superior heat transfer efficiency compared to the other designs.
- **Thermal Gradient**: The hexagonal design maintains a higher thermal gradient, enhancing the overall cooling performance.
- **Heat Flux**: The hexagonal shaped twisted design achieves a higher heat flux, contributing to more effective heat dissipation.
- **Overall Performance**: Due to its better heat transfer capabilities and effective management of thermal gradients, the hexagonal shaped twisted capillary tube is the most efficient choice among the four designs.

# Limitations

- **Manufacturing Complexity**: The hexagonal shaped twisted design may be more challenging and costly to manufacture compared to simpler shapes like the straight tube.
- **Material Stress**: The twisting and unique shape might introduce additional mechanical stresses, potentially affecting the material's durability and lifespan.
- **Pressure Drop**: The complex internal geometry could lead to a higher pressure drop, impacting the efficiency of refrigerant flow.
- **Scalability**: Scaling up the production of hexagonal shaped twisted tubes could require specialized equipment and processes, potentially limiting its widespread adoption.
- **Compatibility**: The performance advantages might vary depending on the specific type of refrigerant used, and not all refrigerants may be equally compatible with the hexagonal design

# References


1. Ashraf Mimi Elsaid, Mohammed Ammar, Ashraf Lashin, and Ghazy M.R. Assassa, “Performance Characteristics of shell and helically coiled tube heat exchanger under different tube cross-sections, inclination angles, and nanofluids”, Case Studies in Thermal Engineering, Vol.: 49, 2023, pp: 103239.
2. Ahmad Zarei, Sadegh Seddighi, Sohail Elahi, and Ramis Orlu, “Experimental investigation of the heat transfer from the helical coil heat exchanger using bubble injection for cold thermal energy storage system”, Applied Thermal Engineering, Vol.: 200, 2022, pp: 117559.
3. Kamel Sigar Hmood, Horatiu Pop, Valentine Apostol, Saleh Al Qaisy, and Viorel Badescu, “Steady-state performance of capillary tubes for small-scale vapour compression systems using different refrigerants”, International Journal of Refrigeration, Vol.:130, 2021, pp: 87-98.
4. Mokhtari Ardekani A, Kalantar V, and Heyhat M.M, “Experimental study on heat transfer enhancement of nanofluid flow through helical tubes”, Advanced Powder Technology, Vol.: 30, 2019, pp: 1815–1822.
5. Dubba Santhosh Kumar and Ravi Kumar, “Adiabatic flow characteristics of R-600a inside a helically coiled capillary tube”, Applied Thermal Engineering, Vol.: 132, 2018, pp: 500–507.
6. Pravin Jadhav, Neeraj Agrawal, and Omprakash Patil, “Flow characteristics of helical capillary tube for transcritical $CO_2$ refrigerant flow”, Energy Procedia, Vol.: 109, 2017, pp: 431-438.
7. Andrew Michael Fsadni, and Justin P.M Whitty, “A review on the two-phase pressure drop characteristics in helically coiled tubes”, Applied Thermal Engineering, Volume 103, 25 June 2016, Pages 616-638
8. Masoud Zareh, Mohammad Ghorbani Heidari, and Puya Javidmand, “Numerical simulation and experimental comparison of the R12, R22 and R134a flow inside straight and coiled helical capillary tubes”, Journal of Mechanical Science and Technology, Vol.: 30(3), 2016, pp: 1421-1430.
9. Hossien Shokouhmand, and Masoud Zareh, “Experimental investigation and numerical simulation of chocked refrigerant flow through helical adiabatic capillary tube”, Applied Thermal Engineering, Vol.: 63, 2014, pp: 119-128.
10. Sukkarin Chingulpitak, and Somchai Wongwises, “A comparison of flow characteristics of refrigerants flowing through adiabatic straight and helical capillary tubes”, International Communications in Heat and Mass Transfer, Vol.: 38, 2011, pp: 398-404.
11. Sukkarin Chingulpitak, and Somchai Wongwises, “Effects of coil diameter and pitch on the flow characteristics of alternative refrigerants flowing through adiabatic helical capillary tubes”, International Communications in Heat and Mass Transfer, Vol.: 37, 2010, pp: 1305-1311.
12. Mittal M.K, Ravi Kumar, and Akhilesh Gupta, “An experimental study of the flow of R-407C in an adiabatic helical capillary tube”, International Journal of Refrigeration, Vol.: 33, 2010, pp: 840-847.